THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 08

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** displacement reaction	Na displaces H
2	**(b)** methyl orange	Synthetic indicator
3	**(d)** water of crystallisation	Anhydrous $CuSO_4$ is white
4	**(a)** potassium	Most reactive
5	**(d)** electrolytic reduction	Electrolysis of molten chlorides
6	**(d)** 2	n- and iso-butane
7	**(a)** ethene	Conc. H_2SO_4, 443 K
8	**(a)** Both are correct	Normal 120/80
9	**(c)** hydrotropism	Water stimulus
10	**(a)** Correct	Both
11	**(c)** cross-pollination	Agents: wind, water, insects
12	**(b)** 4	9 : 3 : 3 : 1
13	**(a)** 44	22 pairs
14	**(b)** +2 D	Add algebraically
15	**(a)** cones in the retina	Cones
16	**(a)** Student A is correct	Must melt
17	**(d)** 40 W	I^2R
18	**(a)** compass	Needle
19	**(d)** cotton cloth	Biodegradable
20	**(b)** omnivore	Human

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	False — rancidity is due to oxidation of fats and oils by air.	2
22	False — it happens in daily life too, e.g. antacids neutralise stomach acid and baking soda neutralises a bee sting.	2
23	Yeast cells; human muscle cells (during vigorous exercise)	1+1
24	The shoot grows upward, away from gravity (negative geotropism).	2
25	Stigma, style, ovary	2
26	All tall (Tt).	2
27	False — it is a ratio of two speeds, so it has no unit.	2
28	A stronger magnetic field in that region.	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Displacement: a more reactive element displaces another, e.g. $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$. Double displacement: exchange of ions, e.g. $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$.	1+1+1
30	Salt + hydrogen: $Mg + H_2SO_4 \longrightarrow MgSO_4 + H_2$; hydrogen — burns with a pop sound	1+1+1
31	Reflex: involuntary, quick, controlled by spinal cord; walking: voluntary, controlled by brain (cerebellum) and conscious thought	3
32	Can cause infection if not done properly; are difficult to reverse; misused for sex-selective abortion	3
33	Scattering of light by colloidal particles; sunlight entering a dense forest canopy; beam in a smoke-filled room	3
34	Heaters draw larger currents; separate circuits with suitable fuse ratings prevent overloading	3

35	Bleaching powder: bleaching cotton/linen, disinfecting drinking water, oxidising agent. Baking soda: baking powder, antacid, fire extinguishers.	1½+1½
	OR: (i) $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (ii) $NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$ (iii) $2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$	1+1+1
36	$O_2 \xrightarrow{UV} O + O$; $O + O_2 \longrightarrow O_3$; UV radiation can cause skin cancer, cataract and harm plants; CFCs	1+1+1
	OR: Shields Earth from harmful UV radiation; $O_2 \longrightarrow O + O$ by UV, then $O + O_2 \longrightarrow O_3$	1+2

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) bending of light at the boundary; incident ray, refracted ray, normal in one plane; Snell's law (ii) $n = c/v$ (iii) light from coin bends away from normal at water–air surface; appears to come from a higher point	2+1+2
	OR: (i) $\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{10} - \frac{1}{15} = \frac{1}{30} \Rightarrow v = +30$ cm; $h' = mh = -8$ cm; real, inverted, magnified (ii) object between F and 2F, image beyond 2F (iii) $m = v/u = -2$	3+1+1
38	(i) P alkane, Q alkene, R alkyne (ii) Q and R (iii) $CH_3{-}CH{=}CH_2$ (iv) P	2+1+1+1
	OR: (i) X: ethanol, Y: ethene (ii) $CH_3CH_2OH \xrightarrow[443\,K]{conc.\ H_2SO_4} CH_2{=}CH_2 + H_2O$ (iii) dehydrating agent (iv) $2\,Na + 2\,CH_3CH_2OH \longrightarrow 2\,CH_3CH_2ONa + H_2$	1+2+1+1
39	(i) $V = V_1 + V_2 + V_3 = I(R_1 + R_2 + R_3) \Rightarrow R_s = R_1 + R_2 + R_3$ (ii) $R = 15\ \Omega$, $I = 0.4$ A, $V_1 = 2$ V, $V_2 = 4$ V	2+3
	OR: (i) 4×60×5 = 1.2 kWh; 1×2 = 2 kWh; total 3.2 kWh (ii) 3.2×30×5 = ₹480 (iii) $I = 1000/220 = 4.55$ A (iv) melts on excess current to protect circuit	2+1+1+1
40	(i) mouth → oesophagus → stomach → small intestine → large intestine → anus; glands: salivary, liver, pancreas (ii) rhythmic contraction of the muscles that pushes food forward (iii) absorbs excess water	3+1+1
	OR: (i) to remove existing starch (ii) covered: no blue-black colour; uncovered: blue-black (iii) light is necessary for photosynthesis (iv) to remove chlorophyll	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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